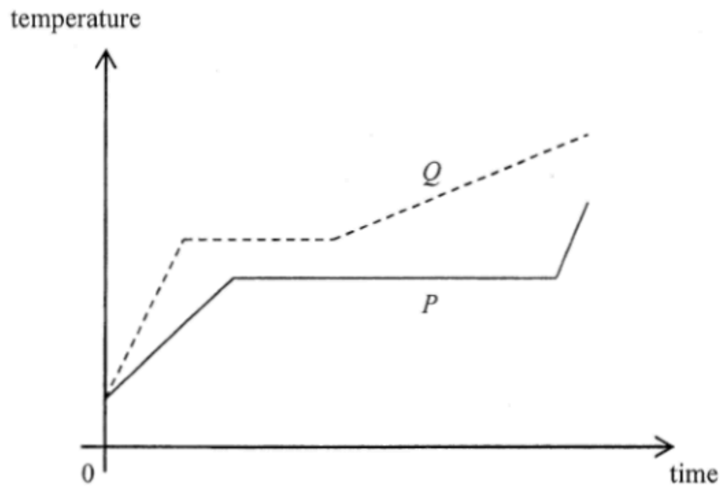
ch3 Change of State

2. Same mass of solids P and Q are heated at the same rate. The temperature-time graphs of the two substances are shown below.



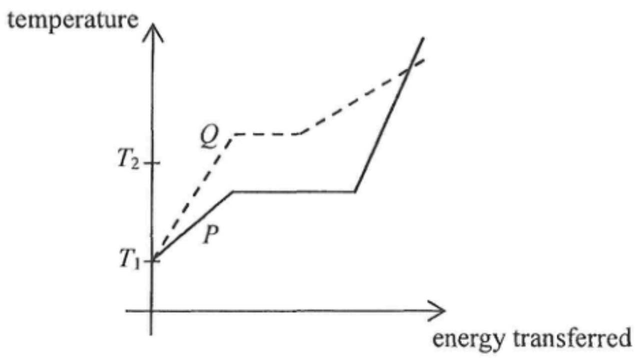
Which of the following comparisons about their melting points and specific latent heats of fusion is correct ?

	higher melting point	**larger specific latent heat of fusion**
A.	P	P
B.	P	Q
C.	Q	P
D.	Q	Q

2. When a patient's arm is wiped by a piece of cotton soaked with alcohol, the wiped area will feel cool as that patch of alcohol on the skin evaporates. Which statement explains this phenomenon ?

A. The evaporation of alcohol absorbs heat from the patient's arm.
B. The alcohol on the skin releases latent heat to the surrounding air.
C. The motion of all the molecules in the patch of alcohol slows down.
D. Air molecules remove heat from the patch of alcohol by conduction.

1. Substance P and Q are of the same mass and both are solids at temperature T_1. The graph below shows the variation of temperature of the substances with energy transferred.



Which statement below is correct ?

A. P has a higher melting point than Q.
B. The specific latent heat of fusion of P is smaller than that of Q.
C. The specific heat capacity of solid P is greater than that of solid Q.
D. At temperature T_2, P is gas while Q is solid.

4. Which of the following descriptions is correct ?

A. When water at 25°C is heated to 50°C, both the kinetic energy and potential energy of the water molecules increase.

B. When water at 25°C is heated to 50°C, only the potential energy of the water molecules increases.

C. When water boils at 100°C and turns into steam, the kinetic energy of the water molecules increases.

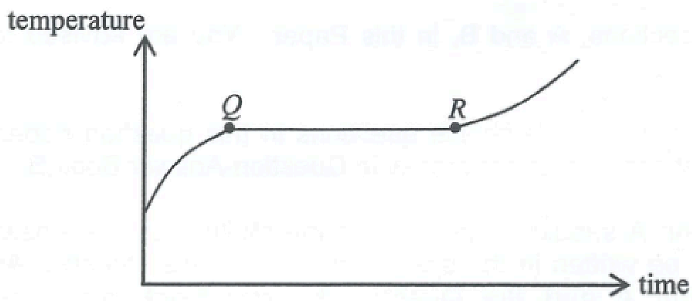
D. When water boils at 100°C and turns into steam, the potential energy of the water molecules increases.

3. Which of the following statements about the internal energy of a substance are correct ?

(1) When a solid melts, the latent heat of fusion absorbed becomes potential energy of the molecules in the substance.
(2) When a vapour condenses, its internal energy decreases.
(3) When a liquid evaporates, the internal energy of the remaining liquid increases.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2.



A substance undergoes the fusion process. The figure shows how the substance's temperature varies with time. Its temperature remains constant during the period from Q to R. Which of the following deductions **within this period** is/are correct ?

(1) The substance does not absorb heat.
(2) The mass ratio of the substance in solid and liquid states remains constant throughout.
(3) The average potential energy of the molecules of the substance increases with time.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2. If 0.5 kg of water at 50 °C is mixed with 2.0 kg of ice at 0 °C, what is the final temperature of the mixture ?

Given: specific heat capacity of water = 4200 J kg^{-1} $°C^{-1}$

specific latent heat of fusion of ice = 3.34×10^5 J kg^{-1}

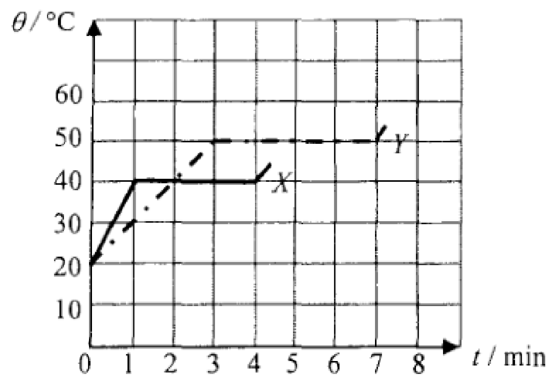
A. –54 °C

B. 0 °C

C. 10 °C

D. 25 °C

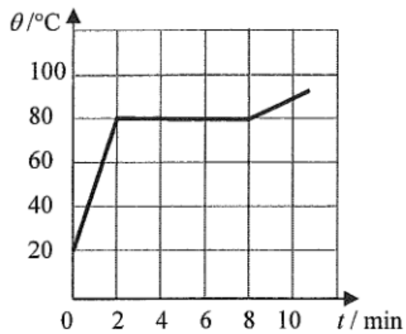
1. Solid substances X and Y of equal mass are heated by heaters of power $2P$ and P respectively. The graph shows how the temperature θ of each substance varies with the heating time t.



What is the ratio of the specific latent heat of fusion of X to that of Y?

A. 3 : 2
B. 3 : 4
C. 4 : 3
D. 2 : 3

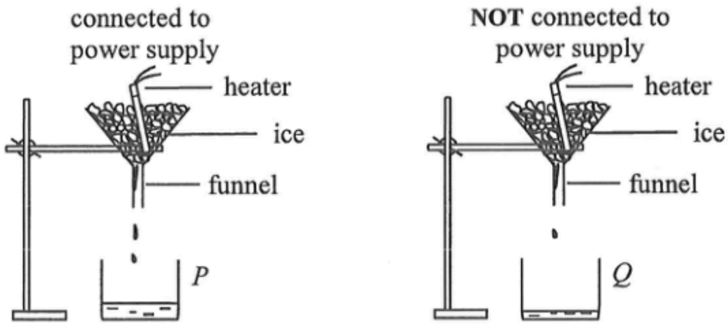
2.



An electric heater of constant power is used to heat a solid substance X which is insulated from the surroundings. The variation of its temperature θ with time t is shown above. X has a specific heat capacity of 800 J kg^{-1} °C^{-1} in its solid state. What is the specific latent heat of fusion of X?

A. 144 kJ kg^{-1}
B. 192 kJ kg^{-1}
C. 202 kJ kg^{-1}
D. Answer cannot be found as both the mass of X and the power of the heater are not known.

2. The set-up below can be used to find the specific latent heat of fusion of ice, l_f. The energy supplied by the heater is measured by a joulemeter. The mass of ice melted by the heater is given by the difference between the mass of water collected in beakers P and Q.



How would the calculated value of l_f be affected if

(I) the heater is not completely covered by the ice ?

(II) the mass of water collected in beaker P is taken as the mass of ice melted by the heater ?

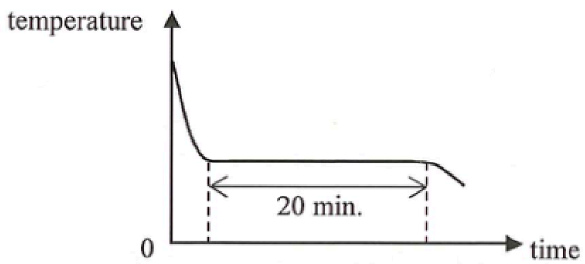
	(I)	(II)
A.	decrease	increase
B.	decrease	decrease
C.	increase	increase
D.	increase	decrease

2. When ice melts into water at 0 °C, what happens to the molecules during the melting process ?

(1) their average separation increases
(2) their average kinetic energy increases
(3) their average potential energy increases

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2.



A hot liquid substance of mass 0.3 kg and with heat capacity 600 J °C^{-1} is left to cool in air. Its temperature falls at a rate of 2 °C per minute just before it begins to solidify. Then its temperature remains steady for 20 minutes until all the liquid just solidifies. Estimate the specific latent heat of fusion of the substance.

A. 20000 J kg^{-1}
B. 24000 J kg^{-1}
C. 48000 J kg^{-1}
D. 80000 J kg^{-1}

2. In an experiment to measure the specific latent heat of vaporization of water, a beaker of water is boiled off using an electric heater. Which of the following sources of error would lead to an experimental result smaller than the standard value ?

A. Energy is lost to the surroundings.
B. Water splashes out of the beaker.
C. Steam condenses on the cooler part of the heater and drops back to the beaker.
D. The heater is not completely immersed in water.

2. 0.3 kg of water at temperature 50 °C is mixed with 0.2 kg of ice at temperature 0 °C in an insulated container of negligible heat capacity. What is the final temperature of the mixture ?

Given: specific heat capacity of water = 4200 J kg^{-1} $°C^{-1}$

specific latent heat of fusion of ice = 3.34×10^5 J kg^{-1}

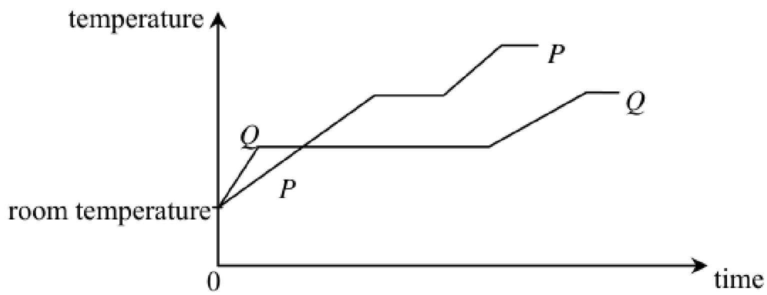
A. −1.8 °C
B. 0 °C
C. 1.8 °C
D. 3.0 °C

1. Which of the following statements about *boiling* and *evaporation* of a liquid is/are correct ?

(1) A liquid absorbs energy when it boils but does not absorb energy when it evaporates.
(2) Boiling occurs at a definite temperature while evaporation takes place above room temperature.
(3) Boiling occurs throughout the liquid while evaporation only takes place at the liquid's surface.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2.



The graph shows the variation in temperature of equal masses of two substances P and Q when they are separately heated by identical heaters. Which deduction is correct ?

A. The melting point of P is lower than that of Q.
B. The specific heat capacity of P in solid state is larger than that of Q.
C. The specific latent heat of fusion of P is larger than that of Q.
D. The energy required to raise the temperature of P from room temperature to boiling point is more than that of Q.